

SmartDrive Assist – Business Plan (3 Phases)

Overview

SmartDrive Assist is a modular conversion kit that transforms a manual wheelchair into a smart, motor-assisted wheelchair. It is designed to be affordable, scalable, and upgradable in phases, making advanced mobility accessible in low- and middle-income regions.

Phase 1: ESSENTIALS (MVP – Core Mobility)

Objective

Provide basic motorized mobility with safety features at the lowest possible cost.

Features

- Manual > Electric conversion kit
- Speed control
- Direction control (joystick/basic controller)
- Brake control (electronic + manual)
- Edge detection (prevent falling from stairs)

- Obstruction detection (avoid collisions)

Components Used

- DC Geared Motors (2x)
- Motor Driver (L298N or similar)
- Microcontroller (basic – Arduino Nano/Uno)
- Ultrasonic Sensors (2–3 for obstacle detection)
- IR Sensors (for edge detection)
- Battery Pack (12V Lithium-ion)
- Joystick Module / Basic Controller
- Wheels + Mounting Hardware
- Wiring + Chassis fittings

❓ Estimated Cost (India)

Component Category	Cost (₹)
Motors + Driver	4,000
Sensors	1,000

Microcontroller	800
Battery	3,000
Mechanical + Frame	3,500
Misc (wires, assembly)	1,200
Total	₹13,500 (~\$160)

Value Proposition

- Affordable mobility solution
- Easy retrofit for existing wheelchairs
- Essential safety features

Phase 2: SECURE+ (Connected & Safe Mobility)

Objective

Add safety, connectivity, and independence features.

Features

Includes everything from Phase 1, plus:

- GPS tracking (real-time location)
- SOS emergency alert (to family/caregivers)
- Bluetooth control (via smartphone)
- Self-charging support (dock or solar assist concept)

Additional Components

- GPS Module (Neo-6M or similar)
- GSM Module (SIM800/900 for SOS alerts)
- Bluetooth Module (HC-05 / built-in if using ESP32)
- Charging Module / Docking Circuit
- Upgraded Microcontroller (ESP32 recommended)
- Mobile App Integration (software)

Estimated Cost (India)

Component Category	Cost (₹)
Phase 1 Cost	13,500

GPS Module	800
GSM Module	1,200
ESP32 (WiFi + Bluetooth)	600
Charging System	2,000
App Development (basic)	2,500
Total	₹20,600 (~\$245)

Value Proposition

- Real-time safety tracking
- Emergency response system
- Remote control via phone
- Increased independence for users

Phase 3: INTELLIGENT (AI-Powered Mobility)

Objective

Deliver a smart, semi-autonomous wheelchair system using AI.

Features

Includes everything from Phase 1 + Phase 2, plus:

- AI-based object recognition (detect people, obstacles, pathways)
- Touch-based controls (capacitive panel)
- Smart navigation assistance
- Voice control (basic commands)
- Path suggestion (assistive navigation)
- Fall-risk prediction (based on terrain detection)

Additional Components

- AI Camera Module (e.g., HuskyLens / Pi Camera)
- Raspberry Pi / Edge AI Processor
- Touch Interface Module
- Microphone (for voice input)
- Speaker / Audio feedback system
- Advanced battery management system

Estimated Cost (India)

Component Category	Cost (₹)
Phase 2 Cost	20,600
AI Camera	5,000
Raspberry Pi / AI Board	6,000
Touch + Audio System	2,000
Advanced Power System	3,000
Software/AI Development	5,000
Total	₹41,600 (~\$500)

Advanced AI Features

- “Follow Me” mode (tracks caregiver)
- Smart indoor navigation (hospital/home use)
- Voice assistant (“Go forward”, “Stop”)

- Obstacle classification (person vs wall vs stair)
- Usage analytics for healthcare monitoring

Business Model Strategy

Phased Product Strategy

- Phase 1 as base product (mass market)
- Offer Phase 2 & 3 as upgrades
- Keep system modular for easy installation

Target Market

- Low & middle-income users
- Hospitals & rehabilitation centers
- Elderly care segment

Revenue Streams

- Hardware sales
- Upgrade modules
- Mobile app premium features (Phase 2+)

Final Vision

A scalable, affordable smart mobility ecosystem that evolves from:

Basic mobility > Connected safety > Intelligent independence